Based on the provided code, answer the following multiple-choice questions:

1. **Given the following code:**

```
LectureCourse lecCourse = new LectureCourse("CS101", "Intro to Programming", "Dr. Smith", 3);
System.out.println(lecCourse.calculateTotalWorkload());
```

hours/week

1 semester = 14 weeks

What will be the output?

3 x 14 = 42 hours

a) 3

b) 14

c) 42

d) 17

2. **Given the following code:**

courseCode Title instructorName numOfSessions

```
LabCourse labCourse = new LabCourse("CS201", "Data Structures Lab", "Prof. Jones", 2.5, 2);
System.out.println(labCourse.getCourseCode());
```

labHours

What will be the output?

a) CS201
b) Data Structures Lab
c) Prof. Jones
d) 2.5

3. **What will be the output of the following code snippet?**

```
LectureCourse lec1 = new LectureCourse("CS101", "Intro to Programming", "Dr. Smith", 3);
LectureCourse lec2 = new LectureCourse("CS102", "OOP", "Prof. Lee", 4);
System.out.println(lec2.calculateTotalWorkload());
```

4 hours per week

∴ 4 x 14 = 56

a) 56
b) 42
c) 3
d) 4

4. **Given the following code, what is the expected output?**

```
LabCourse lab1 = new LabCourse("CS201", "Data Structures Lab", "Prof. Jones", 2.5, 2);
LabCourse lab2 = new LabCourse("CS202", "Algorithms Lab", "Dr. Brown", 2.0, 1);
System.out.println(lab1.getCourseTitle());
```

a) CS201
b) Data Structures Lab
c) Prof. Jones
d) 2.5

5. **Considering the `LectureCourse` class, what would be the result of the following code?**

```
LectureCourse lec = new LectureCourse("CS101", "Intro to Programming", "Dr. Smith", 3);
lec.printCourseDetails();
```

a) A compilation error.
b) A runtime error.
c) The details of the lecture course will be printed to the console.
d) Nothing will be printed.

6. **What is the primary difference between how a `LectureCourse` and a `LabCourse` calculate their total workload?**

a) `LabCourse` uses hours per week * sessions, while `LectureCourse` uses hours per week.
b) `LabCourse` uses a `double` for hours, while `LectureCourse` uses an `int`.
c) `LectureCourse` uses weeks in a semester, while `LabCourse` does not.
d) `LabCourse` does not have a `calculateTotalWorkload()` method.

7. **If we add a new type of course, `OnlineCourse`, which has `videoHours` and `quizCount` as attributes, what method from the `Course` interface would need to be implemented in a unique way for this new class?**

a) `getCourseCode()`
b) `getCourseTitle()`
c) `calculateTotalWorkload()`
d) `printCourseDetails()`

8. **What is the purpose of the `@Override` annotation used in the `LectureCourse` class?**

a) It tells the compiler to ignore the method.
b) It tells the compiler that this method is intended to override a method in a superclass or interface.
c) It is required for all methods in an interface.
d) It is used to define a new method.